

The C++ programming language

In 1979, Danish programmer Bjarne Stroustrup (b. 1950) started working at Bell Labs. He had previously worked using Simula67, considered to be the first object-oriented programming language. Simula67 had been designed to let people model real-world systems easily, but Stroustrup found it quite slow. He decided to add object-oriented features to C, to create a fast language for building large systems. This resulted in C++, which was released in 1983.

```
#include <iostream>

int main(int argc, const char * argv [])
{
    std::cout << "Hello, World!\n";
    return 0;
}
```

C++

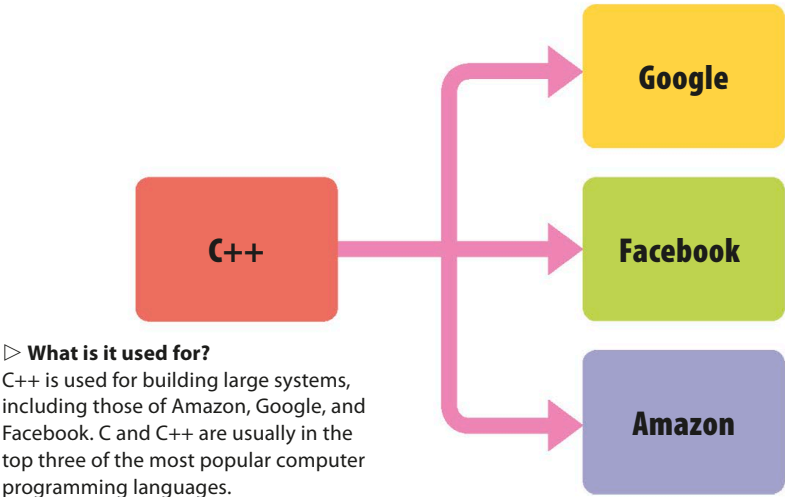
Compiling program for Xcode IDE

Run Without Building	⌘ ⌥ R
Test Without Building	⌘ ⌥ U
Profile Without Building	⌘ ⌥ I
Test	⌘ ⌥ ⌘ R
Test Again	⌘ ⌥ ⌘ R
Profile	
Profile Again	
Compile "main.cpp"	⌘ ⌥ R
Analyze "main.cpp"	⌘ ⌥ ⌘ R

◀ **Compiled**
Similar to C, C++ is compiled to create an executable file before it is run. This can be done on the command line, or through an IDE, such as Visual Studio or Xcode.

How does it work?

C++ looks very similar to C. It also allows programmers to access the computer's hardware in the same way. However, unlike C, it includes features that allow programmers to abstract away from the hardware of the computer without slowing down their code. For instance, data structures are ways of organizing data in a program. C++ includes built-in data structures, whereas C programmers have to code these themselves.



REAL WORLD

C++ at the movies

Autodesk's Maya animation tool is written in C++. Maya has been used to create visual effects for many popular films, including *Star Wars Episode I*, *Spider-Man*, *Lord of the Rings*, and several *Harry Potter* movies. It's possible for programmers to write their own plug-ins in C++ to add functionality to Maya.

C++